

The charger says "faulted". *We say why.*

A small module on the DIN rail that watches the 24 V control supply inside a DC fast-charger cabinet — the layer every relay, contactor and controller depends on, and the one layer OCPP can't see.

< 2 μ shardware transient
detection

12–62 V

12 / 24 / 48 V
systems, one device

NFC

read the last events
at the cabinet, no
app

1,000

events stored on
board, offline

THE BLIND SPOT

Your monitoring stops one layer above the fault.

When a contactor opens, the control rail rings. The controller browns out before it can write a log entry, the dashboard reports "faulted", and the technician drives out to power-cycle a charger that will fail again next week. Nothing in the OCPP data, the backend or the firmware log ever shows the cause.

WHAT RS-24 CHANGES

The spike, the time, the trend — recorded in hardware.

RS-24 triggers on the transient itself, captures its amplitude and envelope, timestamps it against a battery-backed clock and stores it locally before anything else in the cabinet has noticed. Over weeks it trends the rail, the load and the cabinet temperature, so the visit is planned instead of emergency.

WHAT IT DOES

Catches the event

Hardware comparator, 16-bit peak capture and a 50 ms envelope of every transient — timestamped, even through a power loss.

< 2 μ s trigger · 16-bit peak ·
1,000-event buffer

Answers at the cabinet

Tap a phone on the enclosure and read the last events before you open the door. No app, no network, no login. Works in metal cabinets via an external antenna.

Passive NFC · works unpowered

Shows it coming

Rail voltage, current, temperature, vibration and sound trended per device and across the fleet. Alerts on drift before the charger goes offline.

± 0.15 V · fleet dashboard · MQTT / REST

WHO IT'S FOR

CPO FLEET

Know which charger is drifting before it goes offline.

One avoided callout pays for a year of monitoring. Every event is evidence your PSU supplier can't wave away.

SERVICE COMPANY

Arrive with the answer, not the question.

Tap the cabinet, see the fault history, fix it once. Hand over each commissioning with a signed rail-health report.

CHARGER OEM

Turn "no fault found" into field data.

Integrate the bare module at PCB level. Every shipped unit reports real operating conditions; warranty disputes end on data.

VARIANTS

RS-24-S

Standard

DIN rail, IP40, Ethernet / WiFi.
Managed sites.

RS-24-F

Field

IP67, NB-IoT / LTE-M, external NFC antenna. Remote sites.

RS-24-M

OEM module

Bare 65 × 55 mm board, REST API, signed OTA. Factory integration.

Pilot programme 2026

A limited number of pilot slots for CPOs, service companies and OEMs. Units at cost, revenue share on the software, no long-term commitment. You get the instrument and the data; we get the field experience that makes the product right.

Pilots hello@morgen-tech.com

OEM sales@morgen-tech.com

Documentation info@morgen-tech.com

Web morgen-tech.com/railsentinel